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NPS/003/015 – Technical Specification for 33kV, 66kV and 132kV Post Insulators

1. Purpose

This document is the technical specification for 33kV, 66kV and 132kV post insulators for use in CE Electric UK substations.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NPS/003/015	1	Technical Specification For 33kv, 66kv And 132kv Post Insulators

2. Scope

This specification details the requirements for 33kV, 66kV and 132kV post insulators for use in open terminal substations on the CE Electric UK network. It will be necessary to consider and include any project specific requirements as detailed in Appendix 3, Addendum To Supplier Requirements.

3. Technical Requirements

3.1 General Design Requirements

Station post insulators shall comply with the requirements of BS 3297-2 designation “C” and Appendix 1 of this specification.

They may be manufactured from either a grey Polymeric Silicon Rubber material that has been tested in accordance with IEC 62231 and IEC 62217 or a grey porcelain material that has been tested in accordance with IEC 60168. (The CEEUK preference is to move towards the Polymeric Silicon Rubber material.)

Appendix 2 contains a declaration of performance which must be completed for each post type offered. Appendix 3 contains a Addendum to Supplier Requirements which lists any project specific requirements Appendix 4 contains a self certification declaration which must be completed for each insulator type.

3.2 Mechanical ratings

Mechanical ratings for Station Post Insulators have been selected to be compliant with the design methodology stated in BS7354 clause 3.2. Appendix 1 details the resultant values. Manufactures shall confirm compliance with these requirements.

3.3 Creepage Requirements

Creepage requirements for Station Post insulators shall be compliant with IEC 60815 Class III.

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4. References

4.1 External Documentation

Reference	Title
BS 3297-2 1993	Characteristics of indoor and outdoor post insulators for systems with nominal voltages greater than 1000V
BS 7354:1990	Code of practice. Design of high-voltage open terminal substations.
IEC 60815	Guide for the selection of insulators in regard of polluted conditions.
IEC 62231	Composite Station Post insulators for substations with a.c. voltages >1000V up to 245kV - – general definitions, test methods and acceptance criteria.
IEC 62217	Polymeric Insulators for indoor and outdoor use with a nominal voltage >1000V – general definitions, test methods and acceptance criteria.

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G.Hammel	Overhead Line Standards Manager	G.Hammel	30/05/2007

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G.Bolden	Purchasing Manager	G.Bolden	01/06/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
K.Midgeley	Standards Engineer	K.Midgeley	05/06/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S.Johnson	15/06/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	01/06/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	01/06/07

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Appendix 1: Technical specification for 33kV, 66kV and 132kV post insulators for use in open terminal substations

System Voltage	Unit	33kV	66kV	132kV
Rated lightning impulse withstand voltage	kV	200	325	650
Rated short duration power frequency withstand voltage	kV	90	140	275
Pollution Class		Class 3 Heavy	Class 3 Heavy	Class 3 Heavy
Minimum phase to earth creepage distance Dm<300mm	mm	950	1600	N/a
Minimum phase to earth creepage distance Dm>300mm	mm	N/a	N/a	3350
Minimum length	mm	445	770	1500
Minimum cantilever failing load	kN	4.0	4.0	4.0
Minimum tensile failing load	kN	State	State	State
Minimum torsional failing load	kNm	1.2	2.0	3.0
Maximum operational dead weight of conductors (inc. ice and connectors)	kg	40	60	120
Fixing arrangement				
Pitch circle diameter	mm	76	127	127
Thread size		M12	M16	M16
Metal fittings (internal/external)		External	External	External

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Appendix 2: Declaration of technical specification for 33kV, 66kV and 132kV post insulators for use in open terminal substations

System Voltage	Unit	33kV	66kV	132kV
Manufacturer				
Type reference				
Pollution class				
Insulations levels				
Rated lightning impulse withstand voltage	kV			
Rated short duration power frequency withstand voltage	kV			
Minimum phase to earth creepage distance Dm<300mm	mm			N/a
Minimum phase to earth creepage distance Dm>300mm	mm	N/a	N/a	
Insulator length	mm			
Minimum cantilever failing load	kN			
Minimum tensile failing load	kN			
Minimum torsional failing load	kNm			
Maximum operational dead weight of conductors (inc. ice and connectors)	kg			
Fixing arrangement				
Pitch circle diameter	mm			
Thread size				
Metal fittings (internal/external?)				

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Appendix 3: Addendum to supplier requirements

Project specific and installation requirements will be provided by Primary Engineering Projects to be included in this appendix.

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APPENDIX 4 **SELF CERTIFICATION CONFORMANCE DECLARATION**

Station Post insulators shall comply with the latest issues of the relevant international and British Standards.

This check sheet identifies the clauses in IEC 60168, IEC 62231 and IEC 62217 and the clauses of the aforementioned Standards relevant to Station Post insulators for use on CE Electric UK Networks.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or IEC' as appropriate

Manufacturer:

Product Reference:

Name:

Signature:

Date:

Appendix B Tables 1 to 6 follow:

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Table 1

IEC 60168 – Type / sample tests – Mechanical (Ceramic)			IEC 62231 (Composite Insulators)			Remarks
Clause / Sub-clause	Requirement	Conformance Code	Clause / Sub-clause	Requirement	Conformance Code	
5.1	Verification of dimensions		9.1	Verification of dimensions		
5.1.1	General requirements					
5.1.2	Creepage Distance tolerances					
5.1.3	Special tolerances					
5.1.4	Shed Angle					
5.2	Mechanical failing load test					
5.2.1	General requirements					
5.2.2	Mounting arrangements					
5.2.3	Loading arrangements					
5.2.4	Bending test					
5.2.5	Torsion test		9.3.2	Tension load test		
5.2.6	Tensile test		9.3.3	Compression and buckling		
5.2.7	Compressive test			withstand test		
5.3	Deflection tests					
5.4	Temperature Cycle tests					
5.4.2	Test Applicable to post insulators of ceramic material					
5.6	Porosity test					
5.6.1	Test procedure					
5.7	Galvanising test					
5.7.1	Test procedures					

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Table 2

IEC 60168 – Type tests – Electrical (Ceramic)			IEC 62231 – (Composite Insulators)			Remarks
Clause / Sub-clause	Requirement	Confor- mance Code	Clause / Sub-clause	Requirement	Confor- mance Code	
4.5	Dry Lightning impulse withstand voltage test		9.2.1	Dry Lightning impulse withstand voltage test		
4.6	Dry switching – impulse withstand voltage test		-			
4.6	Wet switching – impulse withstand voltage test		9.2.2	Wet switching – impulse withstand voltage test		
4.8	Wet power frequency withstand voltage test		9.2.3	Wet power frequency withstand voltage test		

Table 3

IEC 60168 – Routine tests – mechanical (Ceramic)			IEC 62231 (Composite Insulators)			Remarks
Clause / Sub-clause	Requirement	Confor- mance Code	Clause / Sub-clause	Requirement	Confor- mance Code	
5.8	Visual inspection		10.2 10.3	Verification of dimensions Galvanising test		
5.9	Mechanical test		10.4	Verification of specified mechanical loads		
4.10.1	Electrical tests					

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Table 4

IEC 62217 – Design tests (Composite Insulators)						
Clause / Sub-clause	Requirement	Conformance Code				
9.2	Tests on interfaces and connections of end fittings					
9.2.6.2	Steep front impulse voltage test					
9.2.6.3	Dry power frequency voltage test					
9.3.3.1	1000 h salt fog test					
9.3.4	Flammability test					
9.4.1	Dye penetration test					
9.4.2	Water Penetration tests					

The tests listed in table 4 are only applicable to composite insulators